

Executive Summary:

This guide compares Snowflake and Databricks as platforms for migrating SAS workflows within the CCW environment. While both are secure, scalable, and CMS-approved, Snowflake offers simpler adoption for SQL-heavy analytics, while Databricks is better suited for advanced data science and machine learning pipelines. This comparison reflects current CCW configurations as of mid-2025.

SAS Migration Platform Comparison: Snowflake vs. Databricks

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Note: This document was adapted from internal content originally prepared for the One PI environment, based on the file QRG-SAS-Migration-Key-Product-Review.docx. One PI-specific references have been removed or generalized, and descriptions have been updated to reflect platform capabilities as implemented in the CCW environment.

Introduction

The Centers for Medicare & Medicaid Services (CMS) is directing organizations currently using SAS Enterprise Guide and related tools to migrate their code to alternative platforms. This Quick Reference Guide is designed to help decision-makers evaluate available options and identify the most suitable solution for their projects. CMS currently supports the migration of SAS workloads to SQL and Python-based solutions on either Snowflake or Databricks.

Organizations using SAS for data analytics, business intelligence, and statistical modeling are increasingly exploring migration to modern platforms such as Snowflake and Databricks. These cloud-native environments offer scalable infrastructure, support for modern programming languages like Python and SQL, and the ability to integrate with open-source libraries and AI tools.

This document compares Snowflake and Databricks as platforms for SAS workload migration, focusing on key factors like user experience, data governance, cost management, and advanced analytics capabilities.

Comparing Snowflake and Databricks

Tool History

The following sections provide a brief history of Snowflake and Databricks.

Snowflake

Snowflake originated as a cloud-based data warehousing platform and has since expanded its capabilities from a database-centric foundation. It now supports robust application development, advanced analytics, and data visualization. Snowflake treats application components similarly to database objects, with the cloud-based database serving as the central hub. This architecture results in what many users see as a unified, streamlined, and integrated analytics environment.

Databricks

Databricks is built on Apache Spark, a platform originally developed to support data science and address the performance limitations of MapReduce. The founders aimed to streamline the development of machine learning pipelines—including data ingestion from disparate sources, cleansing, and large-scale model training with scalable hyperparameter tuning. Unlike traditional database-centric platforms, Databricks treats the database as secondary. Instead, it emphasizes a data lake architecture, where data is ingested into the Databricks Lakehouse and governed through the Unity Catalog.

Over time, the capabilities of Snowflake and Databricks have expanded and begun to converge, with many of the initial differences between them narrowing. However, each platform's origins still shape its current strengths, design philosophy, and typical use cases. While neither is inherently better, organizations can make more effective decisions by aligning their selection with the nature of their typical workloads. The following considerations can help guide that choice:

- If your organization focuses on developing and deploying sophisticated machine learning models that involve complex business logic and require large-scale training, Databricks may be the more suitable choice. Its machine learning capabilities are more mature, offering developers greater control over infrastructure for training and modeling. However, this flexibility comes with added complexity—effective use of Databricks typically requires familiarity with Spark concepts and Python programming.
- If your organization primarily conducts ad hoc analytics, statistical research, or large-scale business intelligence and data visualization, Snowflake is generally easier to adopt. It abstracts away infrastructure management and places SQL at the center of analysis. The need for complex frameworks is minimized, and data integration is simplified by running code directly within the database environment.

It is important to recognize that only a small subset of users have mastered the advanced machine learning techniques required to effectively analyze CCW data. The vast majority continue to focus on basic querying and statistical analysis. Nonetheless, system demand driven by artificial intelligence (AI) workloads has increased significantly—particularly over the past year.

Data Governance and Integration

Databricks manages data and AI assets through the Unity Catalog, which provides a graphical user interface (GUI) to oversee access and governance for data, notebooks, machine learning models, dashboards, and other resources. Access policies can be defined at granular levels for both individuals and groups.

Snowflake offers governance features such as data quality and integrity monitoring using both system-generated and custom-defined metrics. Sensitive columns can be tagged for enhanced oversight, and policy compliance can be monitored via integrated dashboards.

Because CCW data is natively housed in Snowflake, users working in Snowflake can inherit existing data protections and integrations with minimal setup. In contrast, accessing CCW data in Databricks that isn't in the Unity Catalog typically requires users to connect to Snowflake and run a query, since the data is not registered in the Unity Catalog or governed directly within Databricks.

Security and Compliance

Both Snowflake and Databricks meet CMS security and compliance requirements, including FedRAMP High standards. In the CCW environment, security controls—such as Role-Based Access Control (RBAC), encryption, and identity management—are centrally implemented and maintained under the infrastructure contract.

As a result, platform-level security features should not be a primary consideration when selecting a tool for SAS workflow migration. Users can assume a secure, compliant environment on either platform without needing to configure or manage security settings themselves.

Cost Management

Although compute and storage costs are covered by CMS under the CCW infrastructure contract, it is still important to understand how each platform manages resource usage and scalability.

- Snowflake uses a consumption-based model that separates compute from storage. While the platform supports features like auto-suspend and auto-resume to minimize idle costs, these are centrally managed in the CCW environment. Currently, users are not directly billed for compute usage, and usage visibility is limited compared to Databricks. However, this may evolve as Snowflake usage grows.
- Databricks operates on a usage model measured in Databricks Units (DBUs). In the CCW implementation, cluster auto-termination policies are enforced centrally, similar to Snowflake. As a result, cost control at the user level is not a significant burden in practice.

While both platforms support cost-efficient usage through built-in scaling and shutdown capabilities, governance and monitoring are primarily managed at the infrastructure level within CCW.

Snowflake

Snowflake is a cloud-native platform that supports both SQL and Python-based analytics. Through Snowpark, a collection of libraries and a managed execution environment, users can run Python and other languages alongside the Snowflake database. While SQL queries execute directly within the database, Snowpark enables Python code to run within secure compute containers—eliminating the need to move data outside the platform.

Snowsight is Snowflake’s web-based interface that provides an interactive workspace for writing code, managing queries, and visualizing results. Its intuitive design allows analysts and non-technical users to easily execute SQL or Python code and explore data without extensive programming experience.

Key Product Features

Using Python on Snowflake, users can take advantage of a wide range of features that support both analytics and machine learning workflows:

- Write code in both SQL and Python
- Use Snowflake Notebooks to develop and manage machine learning pipelines
- Execute both interactive and non-interactive workloads within Snowflake Worksheets, and visualize results through built-in charts and dashboards
- Collaborate and share worksheets and dashboards with other users
- Build interactive applications using Streamlit, fully integrated within the Snowflake environment
- Boost productivity with Snowflake Copilot, a generative AI assistant (availability may vary based on platform configuration)
- *(Note: Git integration is not currently available in the CCW implementations)*

The following table summarizes the key benefits and constraints associated with Python on Snowflake in the CCW environment.

Table 1: Snowflake – Platform Advantages and Limitations in the CCW Environment

Potential Advantages	Potential Limitations
• Snowflake automatically runs all queries in the database context. This eliminates the need to replicate complex security measures. • Snowflake eliminates the need to create scripts that copy and load data to other locations for further analysis since it is already in the database. • The Snowflake Snowsight interface is simpler and more understandable, especially for people without a computer science background. • Snowflake includes managed access to open-source libraries supporting enhanced data processing and machine learning, including Pandas and PySpark. This solution currently supports over 2,300 packages found in the full Anaconda Library.	• Some technical features, such as the following, are not yet available in the CMS FedRAMP instance of Snowflake: • Snowflake Copilot becomes available in May 2025. • Graphics Processing Unit (GPU) workload allocation features, which are useful for machine learning workloads, become available in June 2025. • Snowflake offers less fine-grained programmatic control over cluster sizing, which can be useful in advanced machine learning applications, such as those performing real-time analytics. • Collaboration on code requires intentional sharing of notebooks with individual users • Github Repo integration is not available in CCW implementation of Snowflake

Databricks

Databricks is a unified analytics platform built on Apache Spark, designed to support large-scale data engineering, machine learning, and analytics workflows. It offers a managed environment that simplifies infrastructure and resource management compared to a standalone Spark setup. Databricks extends beyond traditional Spark capabilities by integrating support for data lakes, collaborative notebooks, and a wide range of machine learning and data engineering tools—making it a comprehensive solution for modern data science and analytics workflows.

Key Product Features

Using Python on Databricks, users can leverage a robust set of features tailored for data science, machine learning, and collaborative analytics:

- Write and execute code in both SQL and Python
- Integrate natively with leading AI and machine learning libraries
- Work within Databricks Notebooks, a familiar environment for collaborative development and experimentation
- Boost productivity using the Databricks AI Assistant, a generative AI feature designed to assist with writing and editing code (capabilities may vary by user and use case)
- Create basic dashboards and visualizations directly within the platform
- Manage source control and versioning through integrated Git support (*not currently available in the CCW implementation of Databricks*)
- Use built-in data quality and testing tools through Expectations, Databricks' native framework for validating data

The following table summarizes the key benefits and constraints associated with Python on Databricks in the CCW environment.

Table 2: Python on Databricks Potential Advantages and Limitations

Potential Advantages	Potential Limitations
• There is superior integration with many data science capabilities, such as MLflow, TensorFlow, and others, allowing efficient development and monitoring of AI models. • Databricks architecture is based on Apache Spark, allowing seamless integration with machine learning frameworks and efficient execution of PySpark workloads, especially on large datasets. • Databricks has integrations with established data vendors, data governance tools, data warehouses, and data lakehouse platforms. • Centralized directory structure for notebooks governed by group membership enables better collaboration on code	• To process CCW data that isn't hosted in Databricks, users must connect to Snowflake, requiring an additional step • Github Repo integration is not available in CCW implementation of Databricks • Snowsight equivalent features of Databricks are not available in CCW implementation; all SQL code has to be run through notebooks with a cluster

Note: Availability of some advanced features—such as MLflow, lakehouse integrations, and AI tooling—may differ in the CCW deployment of Databricks. Please consult current environment documentation or administrators for details.

Tool Overview

Snowflake

Snowflake is a cloud-native platform originally built for scalable data warehousing. It now supports Python and SQL through Snowpark and provides a fully managed experience for analytics, data sharing, and application development. Its strengths lie in simplified architecture, strong security, and seamless integration with SQL-based workflows.

Databricks

Databricks is a unified analytics platform built on Apache Spark, designed for large-scale data engineering, machine learning, and streaming workloads. It supports Python, SQL, and other languages, offering a collaborative notebook interface with tight integration of ML and AI libraries.

Side-by-Side Feature Comparison

Feature Area	Snowflake	Databricks
Primary Language Support	SQL, Python (via Snowpark)	SQL, Python (via Spark APIs & notebooks)

Workload Focus	Ad hoc analytics, statistical research, BI dashboards	ML pipelines, AI model training, distributed data processing
Compute Model	Auto-scaling, auto-suspend/resume, serverless containers	Cluster-based compute with static or auto-scaling options, serverless compute available
Data Access	Data resides natively in Snowflake; secure, governed by default	Requires data ingestion or connection to Snowflake; extra setup may be needed
User Interface	Snowsight (web UI for SQL, Python, dashboards)	Databricks Workspace (collaborative notebooks with SQL and Python)
Machine Learning	Streamlit integration, Snowflake Copilot (May 2025), limited GPU support	Native support for MLflow, TensorFlow, Hugging Face; mature GPU & AI tools
Cost Management	Pay-as-you-go; compute and storage billed separately; usage tracking handled centrally in CCW.	Pay-as-you-go via DBUs; usage and cluster controls are centrally managed in CCW.
Data Governance	Integrated RBAC, column-level security, object tagging, policy auditing	Managed via Unity Catalog; access controls for data and assets exist but are centrally configured in CCW and not a major factor for SAS workflow migration.
Code Versioning ¹	Not available in CCW implementation	Not available in CCW implementation

¹ While both platforms support Git integration in standard environments, this functionality is not currently enabled in CCW.

² Both platforms meet CMS security and FedRAMP High standards. In the CCW environment, these controls are centrally managed and not part of user migration decisions.

Strengths & Limitations

Snowflake: Advantages

- Native SQL and Python execution within the database
- Much of the CCW data currently resides in Snowflake, which also supports direct sharing from other Snowflake environments such as IDR
- Auto-scaling compute and seamless workload management
- Less overhead for teams transitioning from SQL-heavy SAS code

Snowflake: Limitations

- Limited programmatic control over compute clusters (e.g., sizing or GPU allocation)

- Code collaboration and sharing must be configured manually on a per-user basis
- Advanced AI/ML capabilities still maturing (GPU support and Copilot expected mid-2025)

☑ Databricks: Advantages

- Advanced ML support (MLFlow, Spark ML, Hugging Face, etc.)
- Cluster-level control for heavy-duty compute, GPU-enabled workloads
- Broader language support, including R
- Shared directory structure allows for easier collaboration on notebooks and code
- Familiar notebook interface for data scientists
- Rich Spark ecosystem for large-scale data transformations and streaming

✗ Databricks: Limitations

- Requires some familiarity with Spark-based workflows and notebook-driven development
- Users must connect to Snowflake to access some CCW datasets
- Compute clusters generally take longer to start up compared to Snowflake
- Databricks SQL warehouses are not available in the CCW environment

Suitability by Use Case

Use Case	Best Fit	Rationale
Migrating SQL-heavy SAS code	Snowflake	SQL-native platform with minimal rewrite and low infrastructure burden
Migrating statistical models from PROC REG/GLM	Snowflake or Databricks	Both support statistical routines via libraries; Snowflake simpler to adopt
Advanced ML pipelines (PROC FOREST, etc.)	Databricks	Better suited for building, training, and tuning large ML models
Daily batch ETL and large joins	Both	Snowflake simplifies with SQL, Databricks handles more complex pipelines
Dashboard creation & exploratory analysis	Snowflake	Snowsight + Streamlit make dashboarding intuitive
Multi-language collaboration (SQL, Py, R)	Databricks	Notebooks support multiple languages inline with version control
Migrating large SAS workflows	Both	Databricks offers intuitive UI-based orchestration for notebook-driven workflows. Snowflake supports task orchestration through SQL-based control flow objects.

Final Considerations

- Snowflake is best suited for teams familiar with SQL and focused on analytic workloads, particularly where simplicity, integration, and low infrastructure overhead are priorities. It excels in environments where code-based logic and reporting are more common than complex machine learning pipelines.
- Databricks is well suited for data science-intensive projects that require Spark-based processing or custom machine learning workflows with greater flexibility and infrastructure control.
- While Git integration is supported in standard platform deployments, it is not currently enabled in CCW. Despite that, both platforms offer collaborative tools and operate within CCW under a centrally managed security and compliance framework that meets CMS requirements.